

Using Bit Manipulation

Change Theme

Language

Python 3



Problem

Given a string containing lowercase English letters, find and print all the characters that occur more than once. You must use bit manipulation to track the characters that have already appeared.

Input Format

The first line contains a string S

Constraints

$1 \leq |S| \leq 10^5$ S contains only lowercase English letters (a-z)

Output Format

Print all duplicate characters in the order of their first repeated occurrence. If there are no duplicates, print No duplicates.

Sample Input 0

programming

Sample Output 0

r g m

Sample Input 1

greater

Sample Output 1

r e

```
1 # Enter your code here. Read input from STDIN. P
2 s = input().strip()
3 seen = 0
4 duplicate = 0
5 for ch in s:
6     bit = ord(ch) - ord('a')
7     mask = 1 << bit
8     if seen & mask:
9         duplicate = duplicate | mask
10    else:
11        seen = seen | mask
12    printed = 0
13    res = []
14    for ch in s:
15        bit = ord(ch) - ord('a')
16        mask = 1 << bit
17        if duplicate & mask:
18            if not printed & mask:
19                res.append(ch)
20                printed = printed | mask
21    if res:
22        print(*res)
23    else:
24        print("No duplicates")
```

Line: 20 Col: 37



Upload Code as File

Run Code

Submit Code



Test against custom input



Test case 0



Test case 1

Compiler Message

Success

